Agent Harness Research System v3

Complete Retrospective and Architecture Guide

Phase 1: Foundations Through J2.3a

Executive Summary

The Agent Harness Research System began with a simple question:

> How do we move from using an LLM as a chatbot to using an LLM as part of a reliable research system?

The central insight was that research quality depends on far more than model capability. A high-quality research system requires:

- Retrieval
- Evidence management
- Source governance
- Contradiction detection
- Gap identification
- Evaluation
- Regression testing
- Observability

Rather than relying on:

```
Question
↓
Prompt
↓
LLM
↓
Answer
```

the project evolved into:

```
Question
↓
Domain Profile
↓
```

```
Retrieval
↓
Evidence Extraction
↓
Evidence Ranking
↓
Contradiction Detection
↓
Research Gap Detection
↓
Coverage Analysis
↓
Memo Synthesis
↓
Evaluation
↓
Regression Testing
↓
Trace Generation
```

The project's most important achievement is not any individual feature.

It is the creation of a reusable Harness Engineering architecture that can operate across multiple research domains while maintaining traceability, quality controls, evaluation, and regression protection.

The harness has successfully operated across:

AI Infrastructure

- NVIDIA Blackwell
- GB200 NVL72
- Vera Rubin
- Data center power
- Cooling
- Networking

and:

Small Modular Reactors

- Licensing
- Construction
- Economics
- Fuel supply
- Grid integration

without changing the underlying research engine.

Only the domain profile changed.

This is the defining characteristic of a successful Harness Engineering implementation.

Core Concepts and Terminology

Document Loading

The process of ingesting source material into the harness.

Examples:

- PDFs
- Markdown files
- HTML pages
- Downloaded web sources

Purpose:

```
Make source material available for analysis.
```

Chunking

Breaking large documents into smaller sections.

Example:

```
200-page PDF
↓
500 chunks
```

Purpose:

```
Enable scalable retrieval and evidence extraction.
```

Retrieval

Selecting the most relevant chunks for a question.

Instead of:

```
Question
↓
All Documents
```

the harness performs:

```
Question
↓
Relevant Chunks
```

Purpose:

```
Reduce noise and improve relevance.
```

Evidence Extraction

Identifying factual claims within retrieved chunks.

Examples:

```
120 kW rack power
```

```
BWRX-300 = 300 MWe
```

```
HALEU supply shortage
```

Purpose:

```
Convert raw text into structured research findings.
```

Evidence Item

A structured research finding.

Example:

```
{
  "id": "E001",
  "claim": "GB200 NVL72 requires ~120 kW",
  "source": "...",
  "confidence": 0.92
}
```

Purpose:

```
Create reusable research building blocks.
```

Evidence Ranking

Scoring evidence according to:

- relevance
- specificity
- confidence
- source quality

Purpose:

```
Allow stronger evidence to influence conclusions more heavily.
```

Source Grounding

Maintaining traceability from conclusions back to source material.

Purpose:

```
Prevent unsupported claims.
```

Source Weighting

Assigning credibility scores to sources.

Example:

```
NVIDIA Technical Documentation
>
Vendor Marketing Material
>
Forum Discussion
```

Purpose:

```
Improve trustworthiness.
```

Memo Synthesis

Combining evidence into a coherent research report.

Purpose:

```
Transform evidence into usable knowledge.
```

Contradiction Detection

Identifying evidence that makes incompatible claims.

Example:

```
24–36 months
vs
8–12 years
```

Purpose:

```
Expose uncertainty and disagreement.
```

Contradiction Normalization

Determining whether an apparent contradiction is actually valid.

Examples:

Not contradictions:

```
300 GW target
vs
13 GW/year licensing throughput
```

```
Rack power
vs
Power shelf power
```

Purpose:

```
Reduce false positives.
```

Research Gap Detection

Identifying important unanswered questions.

Example:

```
No UPS information found.
```

Purpose:

```
Guide future research.
```

Coverage Analysis

Measuring how thoroughly a topic has been researched.

Example:

```
Power      Strong
Cooling    Moderate
Network    Weak
```

Purpose:

```
Measure completeness.
```

Domain Profile

A configuration describing a research domain.

Examples:

```
ai_data_centers.yaml
smr.yaml
```

Purpose:

```
Separate research engine from domain knowledge.
```

Topic Detection

Determining which research topics are relevant.

Example:

AI Infrastructure:

```
Power
Cooling
```

```
Networking
Operations
```

SMR:

```
Licensing
Construction
Economics
Fuel
```

Entity Extraction

Identifying the subject of a claim.

Examples:

```
GB200 NVL72
```

```
Power Shelf
```

```
BWRX-300
```

Purpose:

```
Enable accurate comparison.
```

Scope Extraction

Determining the level at which a claim applies.

Examples:

```
component
rack
cluster
unit
```

```
fleet
site
```

Purpose:

```
Prevent invalid comparisons.
```

Metric Normalization

Converting measurements into comparable forms.

Example:

```
24 months
↓
24 months

2 years
↓
24 months
```

Purpose:

```
Enable reliable contradiction analysis.
```

Evaluation

Benchmarking system performance against known expectations.

Purpose:

```
Measure quality.
```

Regression Testing

Comparing current benchmark results to historical results.

Purpose:

```
Detect quality improvements or degradations.
```

Design Philosophy

The harness treats the LLM as:

```
A component
```

not:

```
The system
```

The harness itself provides:

- workflow
- governance
- observability
- evaluation
- regression protection
- portability

The model performs reasoning.

The harness provides reliability.

Architecture Evolution

Version 0

```
Question
↓
LLM
↓
Answer
```

Problems:

- no traceability
- no retrieval
- no evaluation
- no reproducibility

Early Harness

```
Question
↓
Documents
↓
Evidence
↓
Memo
```

Solved:

- source grounding
- repeatability

Mid Harness

```
Question
↓
Retrieval
↓
Evidence
↓
Contradictions
↓
Memo
```

Solved:

- scalability
- conflict detection

Current Harness

```
Question
↓
Profile
↓
Retrieval
↓
Evidence
↓
Evidence Ranking
↓
Contradictions
↓
Research Gaps
↓
Coverage Analysis
↓
Memo
↓
Evaluation
↓
Regression
↓
Trace
```

Solved:

- portability
- observability
- quality measurement

Major Inflection Points

Inflection Point 1

Harness vs Prompting

The realization that prompting alone cannot provide reliability.

Inflection Point 2

Domain Profiles

The separation of research engine from domain knowledge.

Inflection Point 3

Web Retrieval (K1)

The transition from static corpora to dynamic research.

Inflection Point 4

Evaluation (J2)

The transition from intuition to measurement.

Inflection Point 5

Regression (J2.3)

The transition from measurement to controlled evolution.

Chronological Evolution

Foundations

Initial Harness

Created:

```
Documents
↓
Evidence
↓
Memo
```

Step A – Observability

Added:

- debug mode
- trace generation
- source reporting

Purpose:

```
Understand system behavior.
```

Step B – Claude Integration

Added:

- planning
- evidence extraction
- synthesis

Purpose:

```
Move from templates to reasoning.
```

Step C – Source Grounding

Added:

```
E001
E002
...
```

Purpose:

```
Trace every conclusion.
```

Step D – Question-Aware Evaluation

Purpose:

```
Measure answers against the actual question.
```

Step E – Regression Foundations

Created benchmark evaluation capability.

Scaling

F1–F3

Added:

- evaluation preservation
- evidence ranking
- chunking

Purpose:

```
Scale research beyond a few documents.
```

G1–G3

Added:

- retrieval
- contradiction detection
- research gaps

Purpose:

```
Improve research quality.
```

H1–H2

Added:

- source weighting
- coverage matrix

Purpose:

```
Measure evidence quality and completeness.
```

Domain Portability

J1.0–J1.5

Added:

- domain profiles
- profile-based topics
- profile-based source weighting
- profile-aware contradictions

Purpose:

```
Make the harness reusable.
```

Web Retrieval

K1.0

Added:

```
Question
↓
Web Search
↓
Download
↓
Evidence
```

Purpose:

```
Allow dynamic research.
```

Contradiction Maturity

J1.6

Added:

- entity extraction
- scope extraction
- contradiction suppression

Purpose:

```
Reduce false contradictions.
```

J1.6.1b

Added:

- rack vs component power distinction
- metric taxonomy

Purpose:

```
Fix GB200 false positives.
```

Evaluation Era

J2.1

Created gold benchmark datasets.

Domains:

- NVIDIA
- SMR

- Contradictions

J2.2

Created automated benchmark execution.

Purpose:

```
Measure quality.
```

J2.2a

Added:

- benchmark validation
- failure diagnostics
- evaluation traces

Purpose:

```
Explain failures.
```

Regression Era

J2.3

Added:

- baseline management
- regression reports
- pass/fail logic

Purpose:

```
Detect changes in quality.
```

J2.3a

Validated:

- synthetic regressions
- real regressions

Result:

```
Regression framework proven.
```

Current Capability Matrix

Capability	Status
Document Loading	✅
Chunking	✅
Retrieval	✅
Web Retrieval	✅
Evidence Extraction	✅
Evidence Ranking	✅
Source Weighting	✅
Source Grounding	✅
Contradiction Detection	✅
Entity Extraction	✅
Scope Extraction	✅
Research Gap Detection	✅
Coverage Analysis	✅
Memo Synthesis	✅
Domain Profiles	✅
Evaluation	✅
Regression Testing	✅
Trace Generation	✅

Lessons Learned

What Worked

- Domain profiles
- Evaluation
- Traceability
- Regression testing

What Failed

- Early contradiction logic
- Scope-insensitive comparisons
- Benchmark brittleness
- Blind quality assessment

Surprises

- Web retrieval mattered more than expected.
- Contradiction quality depended heavily on entity/scope awareness.
- Evaluation became more valuable than adding new features.

Current Known Limitations

CONTRA_006

Still a known contradiction limitation.

Suppression Taxonomy

Some suppressions still classify as:

```
entity_mismatch
```

instead of:

```
scope_mismatch
```

Benchmark Brittleness

Some benchmark failures are benchmark-definition issues rather than harness issues.

Retrieval Gaps

Certain SMR benchmark questions still achieve only partial coverage.

Current Benchmark State

Latest validated benchmark:

```
Overall Score:             ~97%
Fact Coverage:             ~96%
Citation Coverage:         100%
Contradiction Accuracy:    100%
```

Regression framework validated through:

- synthetic degradation
- real no-web-search degradation

Phase 1 Outcome

Phase 1 established:

```
Reliable Research Infrastructure
```

The harness can:

- ingest sources
- retrieve evidence
- reason over evidence
- identify contradictions
- identify gaps

- evaluate itself
- detect regressions

across multiple domains.

Phase 2 Roadmap

J3.0

Retrieval Quality Improvement

Focus:

```
Query Expansion
```

J3.1

Evidence Quality Improvement

J3.2

Retrieval Diversity

J3.3

Source Ranking Improvements

Final Assessment

The project successfully evolved from:

```
Prompt
↓
Answer
```

to:

```
Research System
```

with:

- observability
- portability
- evaluation
- regression protection

The completion of J2.3a marks the end of Phase 1 and the beginning of Phase 2: improving research quality rather than infrastructure.